DBMS NORMAL FORMS - 80 PYQ PRACTICE SET

1NF, 2NF, 3NF and 4NF - with functional dependencies, candidate keys, MVDs, and decomposition prerequisites

Coverage	1NF, 2NF, 3NF, 4NF, BCNF comparisons, FDs, candidate keys, MVDs, lossless join, dependency preservation
Exams represented	GATE, ISRO, DRDO, BARC OCES, UPSC ESE Electronics
Format	80 questions; exam-year label is placed directly under every question
Question wording	Paraphrased for study use rather than reproduced verbatim. Memory-based items are explicitly marked.

Important scope note: SSC JE and RRB JE were not used for the CSE/DBMS pool because the indexed JE papers available for those exams are not CSE database-normalization papers. No questions were invented from those exams. The set therefore uses the exams for which relevant CSE/DBMS normalization material could be verified.

How to use this: Try every question without looking up the source first. For figure-based items, the original paper figure is required; those items are clearly marked below.

Section A - Core normalization PYQs

1. Let an inference system include the rule XY -> YZ. Which Armstrong-style rule is sufficient to derive it from a related pair of dependencies? State the corresponding union/augmentation reasoning.

A) Decomposition only

B) Union / add-right-hand-side reasoning

C) Transitivity only

D) Pseudotransitivity only

[GATE CSE - 2026]

Type: MCQ | Note: Paraphrased from the archived normalization question; exact wording/options may differ.

2. Consider statements about normal forms and decomposition: (i) a 3NF decomposition can be lossless and dependency-preserving, (ii) a 1NF relation need not be in 2NF, (iii) every BCNF relation is in 3NF, and (iv) every 2NF relation is in BCNF. Select the statement set that is correct.

A) (i), (ii), (iii)

B) (i), (iii), (iv)

C) (ii), (iii)

D) (i), (ii), (iv)

[GATE CSE - 2026]

Type: MCQ | Note: Statement-set question; paraphrased for study use.

3. For a relation Team(name, city, owner), the dependencies include name -> city and name -> owner. Consider decomposing it into relations that separate the city and owner information. Which statements about losslessness, BCNF, and dependency preservation are valid?

A) The decomposition is necessarily lossy

B) Both component relations can satisfy BCNF

C) The original dependency can be preserved by the decomposition

D) BCNF is impossible here

[GATE CSE - 2025]

Type: MCQ | Note: Paraphrased from the GATE 2025 Set 1 normalization/decomposition question.

4. A relation is said to be in 1NF. Which statement is necessarily true about its attributes and tuple values?

A) Every determinant is a candidate key

B) Attribute values are atomic / indivisible in the relation model used

C) No transitive dependency can exist

D) Every non-key attribute fully depends on every key

[GATE CSE - 2024]

Type: MCQ | Note: 1NF-focused PYQ; paraphrased.

5. A relation contains exactly two attributes. For arbitrary non-trivial functional dependencies on those attributes, which normal-form statement is guaranteed?

A) It can be outside 2NF

B) It can be outside 3NF but inside 2NF

C) It is always in BCNF

D) It is always in 4NF only

[GATE CSE - 2022]

Type: MCQ | Note: Two-attribute BCNF question; paraphrased.

6. For a relation with functional dependencies P -> QR and RS -> T, which of the following dependencies can be inferred using closure and transitivity?

A) PR -> Q

B) PS -> T

C) P -> T

D) QR -> S

[GATE CSE - 2021]

Type: MCQ | Note: FD-closure question used as a normalization prerequisite; paraphrased.

7. If every non-key attribute is functionally dependent on the primary key, then, based only on the statement given, what is the highest normal form that can be guaranteed?

A) 1NF

B) 2NF

C) 3NF

D) 4NF

[ISRO CSE - 2020]

Type: MCQ | Note: Official ISRO Scientist CS 2020 question, paraphrased.

8. For a relation schema with a stated key and functional dependencies, identify the case in which the relation is in 3NF but not BCNF.

A) Every determinant is a superkey

B) A non-prime attribute determines another non-prime attribute

C) A prime attribute determines a non-prime attribute and the determinant is not a superkey

D) There are no functional dependencies

[GATE CSE - 2020]

Type: MCQ | Note: 3NF-vs-BCNF concept question; paraphrased.

9. Suppose a relation is decomposed into Y(PR) and Z(QRS), with a dependency such as QR -> S. Determine which combination of lossless-join, dependency-preservation, and normal-form properties is satisfied by the decomposition.

A) Lossless and dependency-preserving only

B) Lossless but not dependency-preserving

C) Dependency-preserving but lossy

D) Neither

[GATE CSE - 2019]

Type: MCQ | Note: Decomposition question; original options are paraphrased.

10. What does it mean for an attribute set X to be fully functionally dependent on a set Y?

A) X depends on every proper subset of Y

B) X depends on Y and on no proper subset of Y

C) Y depends on X only

D) X and Y must be candidate keys

[ISRO CSE - 2018]

Type: MCQ | Note: Definition of full functional dependency, paraphrased.

11. Consider R(a,b,c,d), where all attribute values are atomic and the only listed non-trivial dependencies are a -> c and b -> d. What is the strongest normal-form conclusion justified by these dependencies?

A) 1NF only

B) 2NF

C) 3NF

D) BCNF

[ISRO CSE - 2018]

Type: MCQ | Note: Archived sources disagree on the keyed interpretation of this item; the question itself is reproduced here only as a study prompt, without an answer key.

12. The original paper presents four relational schemas with their non-trivial functional dependencies and underlined primary keys. Which schema is in 3NF but not BCNF?

A) Schema I

B) Schema II

C) Schema III

D) Schema IV

[GATE CSE - 2018]

Type: Figure-based | Note: Figure-based original item. The source figure is not reproduced in this text compilation, so the four schemas/options require the original paper figure.

13. Given functional dependencies V -> W, VW -> X, Y -> VX, and Y -> Z, reduce them to an irreducible / minimal equivalent set. Which dependency set is the appropriate result?

A) Keep only V -> W and Y -> V

B) Remove redundant attributes and dependencies while preserving the closure

C) Replace all FDs by one superkey dependency

D) Remove Y -> Z because Z is not a key

[GATE CSE - 2017]

Type: MCQ | Note: Minimal-cover / FD-reduction question, paraphrased.

14. The original paper shows a journal/research-article schema and asks for the weakest normal form satisfied by a newer design that was not satisfied by the older design. Identify the normal form.

A) 1NF

B) 2NF

C) 3NF

D) BCNF

[GATE CSE - 2016]

Type: Figure-based | Note: Figure-based original item; the journal schema diagram is not reproduced here.

15. Given FDs PR -> ST and PSU -> QR, determine which dependency is implied / trivial under the stated dependency set.

A) PR -> S

B) PU -> T

C) QS -> P

D) RS -> U

[GATE CSE - 2015]

Type: MCQ | Note: Paraphrased FD-closure/trivial-dependency item; solve from the stated FDs.

16. The original paper supplies a relation instance/table and asks for the highest normal form of that relation. Identify the normal form after checking keys and functional dependencies.

A) 1NF

B) 2NF

C) 3NF

D) BCNF

[ISRO CSE - 2014]

Type: Figure-based | Note: Figure-based original item; the source table is not reproduced in this text compilation.

17. If every value of attribute A always occurs with exactly one B value, while the same A can occur with multiple C values, what dependencies are represented?

A) A -> B and A -> C

B) A -> B and A ->> C

C) A ->> B and A -> C

D) B -> A and C -> A

[ISRO CSE - 2014]

Type: MCQ | Note: Multivalued-dependency question; paraphrased.

18. A relation R is decomposed into R1(A,B) and R2(C,D), with dependencies A -> B and C -> D. Which statement about dependency preservation and lossless join follows from this decomposition?

A) The decomposition preserves the stated dependencies

B) It necessarily destroys all dependencies

C) It is necessarily lossy

D) It violates 1NF

[ISRO CSE - 2014]

Type: MCQ | Note: Decomposition question; paraphrased.

19. What is a prime attribute in a relation schema?

A) An attribute present in every candidate key

B) An attribute that belongs to at least one candidate key

C) An attribute with a unique value in every tuple

D) A foreign-key attribute

[GATE CSE - 2014]

Type: MCQ | Note: Definition question, important for 3NF testing.

20. Consider two statements: S1 claims that a relation containing two attributes is in all standard normal forms; S2 concerns whether a set of functional dependencies is a minimal cover. Select the correct truth-value combination.

A) S1 true, S2 true

B) S1 true, S2 false

C) S1 false, S2 true

D) S1 false, S2 false

[GATE CSE - 2014]

Type: MCQ | Note: Two-statement normalization question; paraphrased.

21. For BOOK(ISBN, Title, Publisher, Address), suppose ISBN -> Title, Publisher and Publisher -> Address. What is the highest normal form guaranteed by these dependencies?

A) 1NF

B) 2NF

C) 3NF

D) BCNF

[ISRO CSE - 2013]

Type: MCQ | Note: Normal-form question, paraphrased.

22. For a relation R(A,B,C,D,E,F,G,H), with atomic attributes and a given set of functional dependencies, determine the highest normal form reached after finding the candidate keys and checking prime/non-prime dependencies.

A) 1NF

B) 2NF

C) 3NF

D) BCNF

[GATE CSE - 2013]

Type: MCQ | Note: The archived question includes a dependency set; this text version states the task without reproducing the full original table/format.

23. For the same relation schema and dependency set, determine the number of candidate keys.

A) 1

B) 2

C) 3

D) 4

[GATE CSE - 2013]

Type: MCQ | Note: Candidate-key prerequisite from the same normalization problem.

24. Which statement correctly compares 3NF and BCNF?

A) Every 3NF relation is BCNF

B) Every BCNF relation is 3NF

C) 3NF is strictly stronger than BCNF

D) 3NF and BCNF are identical

[GATE CSE - 2012]

Type: MCQ | Note: Classic 3NF/BCNF comparison question, paraphrased.

25. Among Armstrong-style inference rules, which dependency relationship is not a primitive inference rule?

A) Reflexivity

B) Augmentation

C) Transitivity

D) Mutual dependency

[ISRO CSE - 2011]

Type: MCQ | Note: Inference-rule question, paraphrased.

26. If every non-key attribute depends fully on a key and the relation otherwise satisfies 1NF, which normal form is obtained from the stated condition?

A) 1NF

B) 2NF

C) 3NF

D) BCNF

[ISRO CSE - 2011]

Type: MCQ | Note: Full functional dependency / 2NF question, paraphrased.

27. Given a Suppliers relation and its functional dependencies, first determine the candidate key(s), then identify the highest normal form satisfied by the relation.

A) 1NF

B) 2NF

C) 3NF

D) BCNF

[GATE CSE - 2009]

Type: MCQ | Note: The original problem is dependency-set based; paraphrased.

28. For R(A,B,C,D,E,P,G) with the functional dependencies supplied in the paper, determine the highest normal form by identifying the candidate key(s) and checking determinants.

A) 1NF

B) 2NF

C) 3NF

D) BCNF

[GATE IT - 2008]

Type: MCQ | Note: Paraphrased from the archived GATE IT 2008 normalization question.

29. For R(A,B,C,D) with A -> B, B -> C, C -> D, and D -> B, analyze the decompositions involving AB, BC, and BD. Determine which statement about lossless join and dependency preservation is correct.

A) All listed decompositions are lossless and dependency-preserving

B) The decomposition is lossless but loses dependency preservation

C) It is dependency-preserving but lossy

D) It is both lossy and not dependency-preserving

[GATE IT - 2008]

Type: MCQ | Note: Decomposition question, paraphrased.

30. For the BOOK/COLLECTION schema from the paper, use the stated dependencies between book, collection, and related attributes to identify the highest normal form.

A) 1NF

B) 2NF

C) 3NF

D) BCNF

[GATE CSE - 2008]

Type: MCQ | Note: Text-archive version; original problem contains the schema/dependency details.

31. BCNF is not generally chosen in which type of situation?

A) When preserving all dependencies is more important than reaching BCNF

B) When the relation has no keys

C) When there are no functional dependencies

D) When all attributes are atomic

[ISRO CSE - 2007]

Type: MCQ | Note: Archived question wording is reported as ambiguous; included as a historical PYQ prompt only.

32. Which dependency relationship is not one of the basic Armstrong inference rules?

A) Reflexivity

B) Augmentation

C) Transitivity

D) Mutual dependency

[ISRO CSE - 2007]

Type: MCQ | Note: Armstrong-rule question; paraphrased.

33. For a set of multivalued-dependency statements, determine how many are implied by the given dependencies using the complementation, replication, coalescence, or transitivity properties of MVDs.

A) 1

B) 2

C) 3

D) 4

[GATE IT - 2007]

Type: MCQ | Note: 4NF-oriented MVD question; paraphrased.

34. Which statement about database normal forms is false?

A) BCNF is stronger than 3NF

B) A relation in BCNF is also in 3NF

C) A relation in 3NF is always in BCNF

D) 2NF removes partial dependencies on candidate keys

[GATE CSE - 2007]

Type: MCQ | Note: Normal-form true/false style question; paraphrased.

35. For a relation with dependencies VY -> W, WX -> Z, and ZY -> V, determine a candidate key.

A) VY

B) WY

C) XY

D) ZY

[GATE IT - 2006]

Type: MCQ | Note: Candidate-key prerequisite from a GATE IT normalization problem.

36. Given A -> B, A -> C, CD -> E, B -> D, and E -> A, identify the dependency that is not implied by the others.

A) A -> B

B) CD -> E

C) B -> D

D) E -> A

[GATE IT - 2005]

Type: MCQ | Note: FD implication question; paraphrased.

37. For R(F1,F2,F3,F4,F5) with F1 -> F3, F2 -> F4, and F1F2 -> F5, identify the highest normal form.

A) 1NF

B) 2NF

C) 3NF

D) BCNF

[GATE IT - 2005]

Type: MCQ | Note: Classic normal-form classification question, paraphrased.

38. Which statement about normal forms and functional dependencies is false?

A) 3NF is weaker than BCNF

B) A relation may be in 3NF but not BCNF

C) Every relation in BCNF is in 3NF

D) Every relation in 2NF is in BCNF

[GATE CSE - 2005]

Type: MCQ | Note: Paraphrased true/false normal-form item.

39. For the Empdtl relation given in the paper, determine the highest normal form after examining its key and the dependencies among employee, department, and related attributes.

A) 1NF

B) 2NF

C) 3NF

D) BCNF

[GATE IT - 2004]

Type: MCQ | Note: Schema-based question; paraphrased.

40. An ER diagram is given and the task is to determine the minimum number of relational tables needed so that the resulting design satisfies 3NF. What is the required minimum?

A) 2

B) 3

C) 4

D) 5

[GATE IT - 2004]

Type: Figure-based | Note: Figure/ER-based original item; source diagram not reproduced here.

41. For the Student Performance schema in the paper, determine its highest normal form from the stated key and dependencies.

A) 1NF

B) 2NF

C) 3NF

D) BCNF

[GATE CSE - 2004]

Type: MCQ | Note: Paraphrased normalization question.

42. Consider a relation with attributes Roll, Name, DOB, and Age, together with dependencies involving Roll, Name, DOB, and Age. Determine the highest normal form.

A) 1NF

B) 2NF

C) 3NF

D) BCNF

[GATE CSE - 2003]

Type: MCQ | Note: Paraphrased schema-classification question.

43. Several decompositions are proposed for a relation with a given FD set. Identify the decomposition that satisfies BCNF rather than merely 3NF.

A) Decomposition I

B) Decomposition II

C) Decomposition III

D) Decomposition IV

[GATE CSE - 2002]

Type: MCQ | Note: BCNF-decomposition identification question; original choices are paraphrased.

44. What is the key effect of converting a relation to BCNF with respect to redundancy caused by functional dependencies?

A) Redundancy can be reduced/eliminated

B) Redundancy must increase

C) Redundancy is unrelated to normalization

D) BCNF removes all storage overhead

[GATE CSE - 2002]

Type: MCQ | Note: Conceptual BCNF question; paraphrased.

45. Given a concrete relation instance and a proposed dependency X -> Y, determine whether the instance satisfies the functional dependency.

A) Yes, for every tuple pair agreeing on X

B) No, because one tuple violates it

C) Only if X is a key

D) Only if the relation is in BCNF

[GATE CSE - 2000]

Type: MCQ | Note: Instance-based FD checking question; paraphrased.

46. A relation is decomposed into two relations whose common attributes form a determinant in the original schema. Determine whether the decomposition achieves 2NF/3NF and whether the join is lossless.

A) Lossless and in 3NF

B) Lossless and only in 2NF

C) Lossy but dependency-preserving

D) Lossy and not dependency-preserving

[GATE CSE - 1999]

Type: MCQ | Note: Decomposition/normal-form question; paraphrased.

47. What normal form is usually considered an adequate target for a sound relational database design in this older GATE question?

A) 1NF

B) 2NF

C) 3NF

D) BCNF

[GATE CSE - 1998]

Type: MCQ | Note: Historical normalization-design question; paraphrased.

48. For R(a,b,c,d), with atomic values and dependencies a -> c and b -> d, determine the strongest normal form that can be guaranteed.

A) 1NF

B) 2NF

C) 3NF

D) BCNF

[GATE CSE - 1997]

Type: MCQ | Note: Historical normal-form classification question, paraphrased.

49. For a relation where BCNF decomposition is being considered, which statement captures the general guarantee concerning lossless join and dependency preservation?

A) Both are always guaranteed simultaneously

B) Lossless join is guaranteed, but dependency preservation may be lost

C) Dependency preservation is guaranteed, but lossless join may be lost

D) Neither can be guaranteed

[GATE CSE - 2001]

Type: MCQ | Note: BCNF decomposition property question; paraphrased.

50. Why can redundancy remain in a relation that is already in 3NF?

A) Because prime attributes can participate in dependencies that do not violate 3NF

B) Because atomicity is not enforced

C) Because every 3NF relation contains multivalued attributes

D) Because BCNF and 3NF are identical

[GATE CSE - 1990]

Type: MCQ | Note: Historical 3NF redundancy question; paraphrased.

Section B - Additional verified normalization / prerequisite PYQs

51. For the relation and dependency set in the paper, determine which attribute set is a candidate key.

A) Set I

B) Set II

C) Set III

D) Set IV

[GATE CSE - 1994]

Type: MCQ | Note: Paraphrased from GATE CSE 1994 question 3.7; the source uses a dependency-based key calculation.

52. Given a proposed decomposition of a relation with its functional dependencies, determine whether the decomposition satisfies BCNF and whether the join is lossless/dependency-preserving.

A) BCNF and lossless

B) BCNF but lossy

C) Not BCNF but lossless

D) Neither

[GATE CSE - 1994]

Type: MCQ | Note: Paraphrased from GATE CSE 1994 question 3.6.

53. Given a relation schema and its dependency set, identify the design that is in 3NF but not BCNF.

A) Design I

B) Design II

C) Design III

D) Design IV

[GATE CSE - 1995]

Type: MCQ | Note: Paraphrased from the archived GATE CSE 1995 normalization question.

54. A "useful" functional dependency X -> Y is one where X and Y are non-empty and disjoint. For a relation with four attributes, how many useful FDs are possible if X and Y are treated as attribute sets?

A) 16

B) 25

C) 50

D) 64

[GATE CSE - 2024]

Type: MCQ | Note: GATE CSE 2024 Set 2 FD-count question; paraphrased.

55. Which implication is valid in general for functional dependencies?

A) If X,Y -> Z,W then X -> Z,W

B) If X -> Z and Y -> W, then X,Y -> Z,W

C) If X -> Z then Z -> X

D) If X -> Y then every subset of Y determines X

[GATE CSE - 2024]

Type: MCQ | Note: GATE CSE 2024 Set 1 FD-implication question; options paraphrased.

56. Consider statements about a dependency-preserving 3NF decomposition and a BCNF decomposition. Which statement about the guaranteed existence of such decompositions is correct?

A) Both dependency-preserving BCNF and dependency-preserving 3NF are always guaranteed

B) A dependency-preserving 3NF decomposition is always obtainable, but dependency preservation is not always guaranteed for BCNF

C) BCNF always preserves every FD by definition

D) 3NF never preserves dependencies

[GATE CSE - 2026]

Type: MCQ | Note: GATE CSE 2026 Set 1 normal-form/decomposition statement question; paraphrased.

57. Suppose the only candidate keys of a relation R(A,B,C,D) are AB and AC. How many superkeys does R have?

A) 2

B) 4

C) 6

D) 8

[GATE CSE - 2026]

Type: MCQ | Note: Superkey-count question; paraphrased.

58. For a stated FD set, identify the attribute set that is NOT contained in the closure of the requested starting set.

A) Option I

B) Option II

C) Option III

D) Option IV

[GATE CSE - 2006]

Type: MCQ | Note: FD-closure prerequisite question from GATE CSE 2006; paraphrased because the source is diagram/table based.

59. For a relation and its FD set, identify all candidate keys.

A) Only AB

B) Only AC

C) AB and AC

D) AB, AC, and BC

[GATE CSE - 2005]

Type: MCQ | Note: Candidate-key question from GATE CSE 2005; paraphrased.

60. A relational schema is given with functional dependencies, and the paper asks whether it is in 3NF and, if not, how it should be decomposed into 3NF. State the correct 3NF decomposition criterion for the given dependency set.

A) Keep every dependency in one table

B) Decompose so that each relevant FD has a key-containing determinant or is represented by a separate relation

C) Remove all non-key attributes

D) Decompose until every relation has one attribute

[GATE CSE - 1993]

Type: MCQ | Note: Paraphrased from GATE CSE 1993 normalization/decomposition question.

61. Given a table and a set of candidate-key conditions, identify whether the design reaches 1NF, 2NF, 3NF, or 4NF. Use the key and dependency definitions rather than the data values alone.

A) 1NF

B) 2NF

C) 3NF

D) 4NF

[ISRO CSE - 2020]

Type: MCQ | Note: Additional ISRO normalization-style PYQ prompt; paraphrased from archived paper sets.

62. Which statement most accurately defines full functional dependency?

A) Dependency on the complete key with no proper subset sufficient

B) Dependency on any superset of a key

C) Dependency only between prime attributes

D) Dependency that implies a multivalued dependency

[ISRO CSE - 2018]

Type: MCQ | Note: ISRO 2018 full-functional-dependency question; paraphrased.

63. For a relation whose candidate key is (Room, Shift), determine which of the listed attribute sets could be another superkey or candidate key after applying closure.

A) Room only

B) Shift only

C) Room + Shift

D) Neither attribute can participate in a key

[ISRO CSE - 2018]

Type: MCQ | Note: Candidate-key question from ISRO CSE 2018, paraphrased.

64. For Faculty(facName, dept, office, rank, dateHired), assume a faculty member has one office and no two faculty in the same department share a name. The relation is in which normal-form combination?

A) Not 3NF, but BCNF

B) 3NF, but not BCNF

C) 3NF and BCNF

D) Neither 3NF nor BCNF

[ISRO CSE - 2017]

Type: MCQ | Note: Official ISRO Scientist CS 2017 Paper II question, paraphrased from the tabular instance.

65. A relation is decomposed into two projections that separate independent functional dependencies. Determine whether the decomposition preserves the original dependencies and whether the join is lossless.

A) Both are preserved

B) Only dependencies are preserved

C) Only lossless join is guaranteed

D) Neither is guaranteed

[ISRO CSE - 2014]

Type: MCQ | Note: Additional ISRO 2014 decomposition prompt, paraphrased.

66. For a table whose dependencies include a non-key determinant, determine whether the relation is in 3NF but not BCNF.

A) Yes

B) No, it is in BCNF

C) No, it is only in 1NF

D) It cannot be decided without atomicity information

[ISRO CSE - 2014]

Type: MCQ | Note: Normalization classification prompt from the archived ISRO 2014 paper set; paraphrased.

67. In BOOK(ISBN, Title, Publisher, Address), given ISBN -> Title, Publisher and Publisher -> Address, which dependency causes the 3NF violation when ISBN is the key?

A) ISBN -> Title

B) ISBN -> Publisher

C) Publisher -> Address

D) Address -> Publisher

[ISRO CSE - 2013]

Type: MCQ | Note: Derived as a focused sub-question from the ISRO 2013 normalization PYQ; paraphrased, not an additional verbatim question.

68. If a non-key attribute is fully functionally dependent on a candidate key, which partial-dependency violation is ruled out?

A) 1NF violation

B) 2NF violation

C) 3NF violation

D) 4NF violation

[ISRO CSE - 2011]

Type: MCQ | Note: Focused PYQ-derived question based on the ISRO 2011 full-dependency item; included as a study variant.

69. Which of the following is not one of the standard Armstrong axioms?

A) Reflexivity

B) Augmentation

C) Transitivity

D) Symmetry

[ISRO CSE - 2011]

Type: MCQ | Note: Armstrong-axiom question from ISRO 2011; paraphrased.

70. Which condition commonly motivates choosing 3NF over BCNF in a relational design?

A) Need to preserve dependencies

B) Need to violate 1NF

C) Need to introduce duplicate rows

D) Need to remove candidate keys

[ISRO CSE - 2007]

Type: MCQ | Note: Normalization-design question associated with ISRO 2007; phrasing is paraphrased.

71. Which inference rule is used to derive a dependency of the form XZ -> YZ from X -> Y?

A) Reflexivity

B) Augmentation

C) Transitivity

D) Union

[ISRO CSE - 2007]

Type: MCQ | Note: Armstrong-rule application, based on the ISRO 2007 question family; paraphrased.

72. What is the role of a foreign key in a relational schema, and why is it relevant to maintaining a well-structured logical design?

A) It guarantees BCNF

B) It enforces referential integrity

C) It removes all transitive dependencies

D) It makes every relation 4NF

[ISRO CSE - 2009]

Type: MCQ | Note: DBMS prerequisite question from ISRO 2009; included because keys/integrity are used when normalizing schemas.

73. Which of the following is a major goal of a good logical relational schema?

A) Increase update anomalies

B) Avoid data inconsistency while supporting queries efficiently

C) Eliminate all keys

D) Store every fact in one large table

[ISRO CSE - 2016]

Type: MCQ | Note: Logical-schema design question from ISRO 2016; included as a normalization-motivation prerequisite.

74. What operation reverses a normalized design back toward a denormalized form?

A) Normalization

B) Denormalization

C) Decomposition

D) Projection

[DRDO CSE - 2022]

Type: MCQ | Note: DRDO CSE 2022 Paper 2, Q15(a), paraphrased.

75. Which of the following are not standard categories of keys in a relational DBMS?

A) Complement and Agreement

B) Candidate and Super key

C) Primary and Foreign key

D) Composite and Candidate key

[DRDO CSE - 2022]

Type: MCQ | Note: DRDO CSE 2022 Paper 2, Q15(c), paraphrased.

76. A four-attribute relation has dependencies A -> B and B -> D. Assuming atomic attributes and no other relevant dependencies are given, what normal-form conclusion can be drawn?

A) 1NF only

B) 2NF only

C) 3NF

D) 4NF

[BARC OCES - 2020]

Type: Memory-based | Note: Memory-based BARC 2020 question report; original wording/data should be checked against the paper archive before treating it as an exact transcription.

77. A table instance is given in which A, B, C, D take several values. Which proposed FD is inconsistent with the instance?

A) AC -> B

B) C -> D

C) B -> D

D) C -> A

[BARC OCES - 2020]

Type: Memory-based | Note: Memory-based BARC 2020 question report; the public archive describes the instance and asks which FD is incorrect. This is a paraphrase, not a verbatim reproduction.

78. Consider these statements about normal forms: (1) every two-attribute relation is BCNF; (2) lossless, dependency-preserving BCNF decomposition is always possible; (3) lossless, dependency-preserving 3NF decomposition is always possible; (4) BCNF is stricter than 3NF. Which statements are correct?

A) 1, 2 and 3

B) 1, 3 and 4

C) 1, 2 and 4

D) 2, 3 and 4

[UPSC ESE Electronics - 2017]

Type: MCQ | Note: Official-paper item identified by Testbook as ESE Electronics 2017; paraphrased.

79. If the only candidate keys of R(A,B,C,D) are AB and AC, how many distinct superkeys exist?

A) 2

B) 4

C) 6

D) 8

[GATE CSE - 2024]

Type: MCQ | Note: GATE CSE 2024 candidate/superkey question; paraphrased.

80. A memory-based report of the BARC 2020 paper records a DBMS question asking candidates to identify which proposed decomposition is invalid with respect to BCNF. What must be checked before accepting a decomposition as a BCNF decomposition?

A) Every determinant in every component must be a superkey of that component

B) Every component must contain exactly two attributes

C) Every FD must be removed

D) Every relation must be in 4NF

[BARC OCES - 2020]

Type: Memory-based | Note: Memory-based BARC 2020 report records this BCNF-decomposition question but does not preserve the original option data; this item therefore tests the underlying criterion rather than pretending to reproduce the lost choices.

Source and verification notes

This is a study compilation, not an official reproduction of any exam paper. Most questions are paraphrased from publicly indexed PYQ pages and exam archives. The exam-year labels are intended to identify the source paper, while the wording has been compressed or rephrased.

Primary verification groups used:

• GATE CSE/IT normalization archive covering decades of normalization-tagged PYQs, cross-checked against solved-paper pages.

• ISRO Scientist CSE/CS solved-paper archives, including 2011, 2013, 2014, 2017, 2018 and 2020 items.

• DRDO CSE 2022 Paper 2 archive for the normalization/denormalization and key questions.

• BARC 2020 Scientist memory-based paper report for the explicitly marked memory-based items.

• UPSC ESE Electronics 2017 official-paper listing for the normal-forms statements question.

Accuracy note: For a few older figure-based or memory-based items, public text archives preserve the question only partially. Those are explicitly labelled rather than silently reconstructed as exact PYQs. Where an original figure is required, the prompt states that the figure is not reproduced here.

Answer key intentionally omitted: The request was for a question-only PYQ set. This keeps the document useful as a practice paper; answers can be prepared separately from the verified source keys.